

SARTHAK PANI

Physics & Astrophysics Student | Technical Founder | AI, Embedded Systems & Computer Vision
Dublin, CA | fficial.sarthakp@gmail.com | [LinkedIn](#) | [GitHub](#) | [Elev](#)

EDUCATION

Diablo Valley College & California Community Colleges

Physics & Astrophysics Transfer Preparation

California

2026-Present

GPA & Honors: 4.00 across 25 transcript-verified Spring 2026 units; DVC Academic Honors.

Completed Math & CS: Calculus III (MATH 3C), Introduction to Linear Algebra (MATH P208), and Introduction to Programming (COMSC 110).

Summer 2026 CS: Python Programming (COMSC 140) and Advanced Programming with C/C++ (COMSC 165).

University of California, Merced

Merced, CA

Lower-Division Mechanical Engineering Coursework

2024-2025

Math, Physics & Engineering: Calculus I-II for Physical Sciences and Engineering (MATH 021/022); Introductory Statistics; calculus-based mechanics and laboratory (PHYS 008/008L); Engineering Computing (ME 021).

Independent AI Coursework: Neural Networks and Deep Learning; Machine Learning; Structuring Machine Learning Projects; Marketing Analytics; Computer Vision and Deep Learning Applications; Mastering OpenCV with Python.

EXPERIENCE

Elev Inc. - Co-Founder & CEO | [Case Study](#) | Dublin / Merced, CA | Remote

2024-Present

- Co-founded and built an AI evidence platform that converts program goals, grants, contracts, staff roles, schedules, and field submissions into tailored collection strategies and source-backed reports.
- Secured a fully paid \$12,000 six-month design pilot with Boys & Girls Club of Merced County and a paid \$1,000 deployment with Merced United; preparing pilots with Merced Parks and Recreation, Youth for Christ, and ARTREE.
- Architected a *React/TypeScript, Node/Express, FastAPI/LangGraph, PostgreSQL/pgvector, Redis, and Azure* system with hybrid retrieval, citations, grounding checks, PII-redacted embeddings, and encrypted chat transcripts.
- First-named inventor on a provisional U.S. patent application filed May 28, 2026, covering AI-generated, grant-aligned program-quality workflows.

Next Generation Marketing, Inc. - Sales Associate | On-site

Jun-Aug 2024

- Generated \$40,000+ in direct product sales over two months, including a personal record of \$13,000 sold in one day.
- Worked across retail locations, adapting product demonstrations, customer conversations, and closing approaches to different audiences.

FutbolForAll - Founder & Nonprofit Director | [Impact](#) | Youth Sports Access

Jun 2022-Jan 2024

- Founded a nonprofit initiative that supplied soccer equipment to children in underserved international communities.
- Recruited and trained a 20-person team, built marketing and fundraising operations, and raised \$1,000+ with support connected to Olympus FC.
- Coordinated a 200+ lb soccer-equipment shipment to Bhalopahar, India - including 31 cleats, 35 balls, 54 shirts, 27 shorts, and 75 additional items - and built community partnerships with Sports Basement and Oakland Roots Sports Club.

Z.Incorp - Founder | E-commerce | Remote

Feb-Dec 2022

- Built a youth e-commerce venture to \$25,000 in aggregate sales and 8 million+ organically reached accounts across TikTok and Instagram.
- Managed product selection, organic content strategy, customer service, inventory, fulfillment, and performance analysis.

SELECTED ENGINEERING PROJECTS

Embedded Robotics & Vision Platform | ESP32, Raspberry Pi, C++, PlatformIO

Personal Project

- Built a Wi-Fi/UDP control link between an ESP32 and Raspberry Pi, integrating L298N motor driving and PWM-based motion control.
- Developed an ESP32-CAM MJPEG web interface for live video streaming and image controls, connecting embedded hardware, networking, and browser-based operation.

Computer Vision Prototypes | Python, OpenCV, pose estimation

Personal Projects

- Built facial-recognition and real-time object detection/tracking prototypes that convert camera input into identity and motion signals.
- Created an AI boxing-coach prototype using pose landmarks and movement analysis to provide feedback on form and technique.

LEADERSHIP & RECOGNITION

- Speaker, [Google Developer Groups DevFest Central Valley 2025](#) - "A Startup Journey in the Central Valley," focused on building and selling an early-stage technology company.
- Selected for [WilmerHale QuickLaunch](#) (\$30,000 in legal-services support) and Microsoft for Startups (\$5,000 in Azure cloud credits).

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript/TypeScript, SQL

Software & AI: React, Node.js/Express, FastAPI, LangGraph, OpenCV, embeddings/RAG, API design, document intelligence

Data & Infrastructure: PostgreSQL, pgvector, Redis, Azure, Docker, Git, Linux

Hardware & Engineering: ESP32, Raspberry Pi, PlatformIO, UDP networking, motor control/PWM, Fusion 360